# A Multi-Stage Deep Learning Framework for Real-Time Plant Leaf Disease Detection and Severity Analysis*


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***Abstract*—Growing rates of crop disease and insect infestations are threatening the overall health of the agricultural industry as well as food production worldwide, and traditional methods for monitoring crops through visual inspection are often too time-consuming, subjective, and difficult to scale up. This paper presents an AI-based system capable of diagnosing crop leaf health in real-time using state-of-the-art computer vision and deep learning technologies. The framework implemented uses a multi-step pipeline composed of an object detection model, a classification model and a vision-based reasoning system; where the object detection portion of the workflow uses a YOLOv8 model to determine the location of infected areas, an automated region-of-interest extraction tool is then used to extract from the image the regions of interest so as to improve the overall accuracy of disease detection. A second EfficientNet based model is then used to estimate the severity of the disease found in the infected area, while an additional vision-based reasoning module is added that looks for visual patterns associated with pest infestations in order to provide contextual information to assist with the diagnosis process. The methodology used to develop this framework includes (1) preprocessing image datasets, (2) training the models on agricultural datasets and (3) deploying an end-to-end inference pipeline that can process user provided inputs through the detection, feature extraction, and diagnosis phases. Preliminary experimental results show high levels of accuracy in the detection of disease and classification of severity, in addition to the implementation of multiple models in a single framework resulting in greater problem-solving capabilities overall.The proposed system provides a scalable and efficient solution for intelligent crop monitoring, enabling early disease detection and supporting data-driven decision making in agricultural practices.**



***Index Terms*—Artificial Intelligence, Plant Disease Detection, YOLOv8, EfficientNet, Computer Vision, Deep Learning, Precision Agriculture**


## I. Introduction

Agriculture is the foundation in providing food security and economic sustainability across the globe, but diseases like plant disease, pests are key factors of crop loss (yield & quality). Good crop management is built on early disease detection and a correct diagnosis of the pathogen involved. Ironically, actual inspection by experts is time-consuming, biased and extremely difficult to scale in large-scale farmland [1], [5]. These restrictions render real time diagnosis an urgent necessity for intelligent and automated systems.

In recent years, artificial intelligence (AI) has made great strides in the fields of computer vision and deep learning, which has resulted in promising progress in image-based plant disease detection. Convolutional neural networks (CNNs) have shown great ability in extracting complex visual features and high classification accuracy in many agricultural datasets [10], [13] . The availability of large annotated datasets and the use of transfer learning techniques have further accelerated the development of automated plant health monitoring systems [11], [12] . However, despite these advances, most of the existing methods are limited to the task of image classification and do not provide spatial localization of infected regions, which reduces interpretability and practical usability in real world scenarios [6], [18].

Real-time localization of regions in images affected by disease has been facilitated by object detection frameworks such as YOLO, improving accuracy and interpretability [7], [8]. However, some existing systems lack combination of detection with advanced diagnosis features such as severity estimation and contextual analysis [2]. In addition, traditional image processing techniques are not robust to complicated environmental conditions, e.g. lighting changes, background

noise and occlusions [20] . These challenges call for the development of more comprehensive and robust diagnostic frameworks.

In this context, the integration of several deep learning models into a single pipeline has become a promising technique to improve the diagnostic performance. More sophisticated architectures like EfficientNet offer better accuracy and computational efficiency for real-time applications [9]. Hybrid models combining convolutional models with advanced learning techniques have shown improved performance for complex recognition tasks [3], [4]. The combination of spatial detection, feature extraction, and contextual reasoning can be utilized to construct a more robust and scalable system for plant disease diagnostics.

An AI system was developed to improve the accuracy and speed of recognising how healthy a plant's leaf is. This system has four main features: an object detector for determining where to find the leaf, a classifier using deep learning to assess how bad the leaf is, and two other methods to help improve diagnostic reliability. Each feature can be used separately or combined for a modular configuration for scalability and field deployment. These features are the result of some new, novel approaches to developing a multi-model diagnostic system, achieving greater accuracy as a result of using multiple algorithms together, and a scalable solution to enable intelligent agricultural decision making.

## II. Literature Review

Many researchers have looked into detecting plant diseases through various ways ranging from traditional approaches involving image processing to applying deep learning methods. The earlier methods focused mainly on handcrafted feature extraction methods where color, texture and shape were the commonly used features, thus allowing researchers to find patterns of disease development based on visible characteristics in plant leaves. Traditional methods provide the ability to process images quickly; however, traditional methodologies do not perform well, due to illumination changes, background interference and large-scale variability characteristics, resulting in poor performance in terms of generalisation [20].

With the advent of deep learning, in particular convolutional neural networks (CNN), the field of plant disease classification tasks has witnessed remarkable progress. Foundational works demonstrated that deep CNN architectures are effective in image recognition, enabling automated feature extraction and higher classification accuracy [10], [16]. The same models were used in later studies on agricultural datasets where promising results were reported in terms of disease identification in various types of crops [11], [13] . Transfer learning was used to boost model performance using pre-trained networks, reducing the dependence on massive labeled datasets [6], [17].

Recent studies are aimed at developing architectures that can increase efficiency and accuracy in existing models. The introduction of compound scaling by EfficientNet have established a new paradigm of optimizing depth as well as width & resolution; leading to superior performance with reduced number of overall parameters for each model [9]. Research using EfficientNet for classification of plant diseases have yielded greater levels of generalization and accuracy than traditional CNN models [3] . Convolutional Neural Network models paired with vision transformers have also shown greater ability to capture local & global features leading to improvements in recognition performance as well [4]. Overall, comprehensive reviews show increased adoption of deep learning methods in agriculture while identifying many opportunities including but not limited to implementation scalability and interpretability [1], [5] issues.

There are a lot of object detection frameworks now that try to fix the problems with models that only do classification. Methods like R-CNN that look at parts of an image have given us the location of objects in images but they took a long time to work [14]. The YOLO framework has changed object detection by giving us image processing and correct detection which makes it useful for real life [7]. The new versions of YOLO like YOLOv5 and YOLOv8 have made object detection faster and more precise, which is really helpful for farmers who want to detect crop diseases in their fields using YOLO. YOLO has really helped farmers use YOLO for crop disease detection, in their fields [2], [8].

Current models have made great progress; however most current models are limited to one function (either detection or classification), and most lack integration of various diagnostic capabilities. Many current models do not include either severity estimation or contextual analysis needed to make effective, well-informed decisions in agricultural settings. Various issues persist in the form of changing conditions, occlusions, and limitations in data sets that continue to affect model performance. As such, a multi-stage, comprehensive framework that includes detection, classification, and higher-order reasoning is necessary for developing reliable, valid, and interpretable plant health diagnosis.

TABLE I
Comparison of Existing Methods

| **Method** | **Technique** | **Limitation** |
|---|---|---|
| Traditional Methods [20] | Color and texture feature extraction | Sensitive to noise and illumination variations |
| CNN-based Models [11], [13] | Deep convolutional neural networks | Lack of spatial or region-level localization |
| EfficientNet Models [3], [9] | Scaled convolutional neural architecture | Limited model interpretability |
| YOLO-based Detection [2], [7], [8] | Real-time object detection framework | Does not estimate severity of detected regions |
| Existing AI Systems [1], [5] | End-to-end deep learning pipelines | Lack of multi-stage analytical capability |

## III. Proposed System Overview

This new system is a multi-stage deep learning system to assist in the diagnosis of plant leaf disease and offers an alternative to the traditional single model approach. Many current models rely on using convolutional neural networks (CNN) to conduct image-level classification; while CNNs are good at classifying images they do not take into consideration

any form of spatial context and struggle when trying to identify localized patterns of disease [6], [18]. There are many examples of deep learning based models reaching a high level of classification accuracy [1], [5], however, as they do not facilitate region-based analysis they may offer limited objectivity and reliability for diagnosis when looking at single images.

The system has been designed to combine detection and identification into one process in order to address the challenges associated with this type of analysis. Using object detection models like YOLO, it is possible to provide real-time identification of areas of interest (i.e., affected) within a given image [7], [8]. In conjunction, using EfficientNet-based architectures for classification [3], [9] can provide a highly accurate representation of the identified areas and therefore combine both spatial and semantic knowledge about the disease-affected plants being analyzed.

Sequential options for processing an input image include preprocessing, detection, region extraction, and classification. Each stage transforms the data representation for the next stage and strengthens the final result's robustness and features. The multi-stage model corresponds to newly developed hybrid deep learning systems with better performance through integration of models [4]. The modular framework for the system provides scalability and deployment to real-world settings (as detailed in Table II).

### A. System Description

The proposed system is an end-to-end pipeline transforming the original input images into structured diagnostic output. At the beginning of the pipeline, the input images will be preprocessed so that they can be normalized and resized to create an image set with consistent dimensions. Once the images are preprocessed and normalized, they will be passed to the real-time object detection model, which will localize infected areas by locating them within the overall image of the leaf. This type of region-aware analysis is the primary shortcoming of most current CNN models, which process whole images without regard to their spatial distributions. [11], [13].

Once the localised areas have been identified; the areas are separated using an automatic crop process that removes noise (from the background). This reduces background noise and improves the relevance of identifying features used in the classification process. The analysis of the areas; is done using an EfficientNet-based deep learning model, which produces predictions for the season type and severity of the diseases. The EfficientNet model uses a compound scaling algorithm that improves accuracy while improving calculation efficiency. [9].

Ultimately the outputs coming from the detection and classification will be combined to form an overall diagnostic output. The overall accuracy of the system is improved by integrating the outputs from several processing stages instead of doing them in a single step approach. The use of both transfer learning and/or pre-trained models can also further enhance its accuracy in conditions where agricultural datasets are scarce. [6], [17].

### B. Key Features

The system suggested here has a multi-stage processing pipeline to implement object detection and classification to manage the entire process of diagnosing plant diseases. With this multi-stage, region-aware analysis, it is possible to locate infected areas accurately and significantly increase the interpretability and accuracy of the diagnosis, unlike conventional image-based classification methods.

The main element of the design is an automatic mechanism to extract regions of interest so as to classify only the areas where disease exists. Thus, greatly reducing the impact of background noise and irrelevant characteristics, which has been a common limitation of most "classic" deep learning systems trained on full images. [11], [13].

This system significantly benefits from the power of contemporary deep learning systems by using two distinct yet complementary parts - YOLO-based real-time object detection [2], [8] and EfficientNet-based classification models [3], [9]. The combination of both provides the capability to achieve both high levels of precision and outstanding computational efficiency in order to enable real-time agricultural applications.

The Independence of each component allows for greater ease in both scaling and updating each component, allowing new technologies and processing methods as they become available (ex. transformers, multimodal processing, and hybrid DL systems) to be adopted into the system because they represent improvements in their respective areas through the course of time. [4].

### C. Novelty

A new unified multi-stage diagnostic framework is proposed that integrates regional detection and classification in order to solve some of the major limitations of current plant disease detection technologies. Unlike traditional methods which classify an entire image and then make a prediction based on this global classification [10], [13], the new method incorporates analysis of specific regions of interest so that predictions are made based on localised characteristics of a disease.

Utilizing an automatic method for isolating (extracting) infected areas from the entire scanned image through region-of-interest extraction significantly improves feature quality (by isolating the infected areas) which provides for better overall classification accuracy and fewer false predictions, especially in more complicated areas with much variability in the background when compared to what would normally occur with non-complex (non-variability) and/or clean (uncontaminated with background clutter).

The framework offers both spatial and semantic capabilities by utilizing a dual-modal based approach with object detection for classification of data. Most of the available YOLO type detection only cover detection [2] capabilities. The proposed expansion will further help to determine disease severity and

improve diagnostic evaluations through the addition of an estimation process to the detection process.

The proposed design ultimately focuses on modularity and the potential for increased scalability, offering the flexibility of adding various types of models to be deployed in real-time. Combining region-aware processing, multi-model integration, and extensible architecture makes our system unique compared to current solutions. This combination provides a more reliable and thorough method for the intelligent diagnosis of diseases in plants.

TABLE II
SYSTEM COMPONENTS

| Module | Description |
|---|---|
| Input Interface | Accepts user image input and manages data flow |
| Preprocessing Module | Performs normalization, resizing, and noise reduction |
| Detection Module | Localizes infected regions using YOLO-based models |
| Region Extraction Module | Extracts relevant regions for focused analysis |
| Classification Module | Predicts disease type and severity using EfficientNet |
| Diagnostic Aggregation Module | Combines outputs from multiple processing stages |
| Output Interface | Displays final diagnostic results |

## IV. SYSTEM ARCHITECTURE

The architecture of this system is built as a systematic multistage pipeline with the ability to combine preprocessing, spatial feature detect, and diagnostic analysis into one complete real time system for the identification of disease in plants. This proposed architecture provides for the combination of the two processes within one workflow, as opposed to most traditional architectures which view detection and classification as independent functions, leading to better diagnostic accuracy and interpretability. By using an object detection system model for localization and a DCNN for classification, the architecture achieves the needed degree of both spatial and semantic understanding of plant disease occurrences. [2], [3], [8].

The architectural design uses a modular approach whereby every module performs a distinct operation and communicates effortlessly with all other modules. Modularity provides the capability to scale up or down to accommodate a variety of different data sets as well as deployment settings. Integration of processing focused on multiple regions of interest (ROI) and multi-model integration offers an alternative to classic convolutional neural networks (CNN), which treat an entire image as one unit without considering spatial context. [6], [18]. The complete structure of the system is illustrated in Fig. 1.

### *A. Architecture Overview*

There are three main levels of architecture: the input/preprocessing level, the detection level and the diagnostic level. The input level has responsibility for obtaining the leaf picture as well as conducting initial preprocessing of that image (e.g., resizing and normalising). This is done to ensure that all of the images being processed have similar appearance when used in model training, in order to provide better consistency in the data when being used, therefore improving the performance of the model.

By using a YOLO-based model for real-time object detection, the detection layer provides spatial awareness to the system and allows it to zone in on the pertinent parts of an image [7], [8] instead of processing the full image as one large block of data. The result is that infected areas are identified and isolated to improve the overall quality of the features that are extracted and reduce the amount of background noise from the image.

The extracted areas go through a classification and analysis process in the diagnostic section. A deep convulutional model based on EfficientNet, with optimised scaling and high prediction accuracy [3], [9], was used to determine how severe the disease is, as well as what type of disease the patient has. An overall diagnostic result is created by merging outputs from different levels. The separate levels work together while being independent of one another and also integrate with one another so that data can flow through all levels efficiently and in real-time.

**Multi-Stage Feature Fusion**

$$F_{fusion} = \lambda F_{YOLO} + (1 - \lambda) F_{EfficientNet} \quad (1)$$

### *B. Workflow Explanation*

A user interface (UI) enables you to take a photo of a leaf. We send the photo to our processing backend server. The photo undergoes preprocessing, which ensures it has the same size and quality. By allowing the model to be tested with different input conditions, we guarantee that these will not introduce variance in the model's ability to predict the expected outcome.

Following a picture being taken by the camera, it will be sent to a machine that can identify diseased portions of the picture with a YOLO model to help locate the infection. This is an important part of making a correct diagnosis. After determining where the infected area is, the machine can then remove the infected portions of the picture. The camera is used to capture an image and the YOLO-based model is used to assist the machine in identifying the infected portion of the image.

The classification module then analyses the extracted regions, predicting the disease category and the degree of severity. For this stage, it uses deep convolutional networks (DCN) to extract high-level features from the region images and provide accurate predictions based on these features. The predicted outputs from both detection/identification (DAI) and classification (CL) form the basis for making a final diagnosis in the diagnostic module, which integrates the two previous steps' results into one single outcome. The processed output is then received by the user's interface as useful means of actionable and informative insight regarding the health of the plants. The structured data processing workflow maximizes efficiency by reducing the occurrence of false predictive outputs,

and therefore increasing the overall reliability of the real-time agricultural systems.

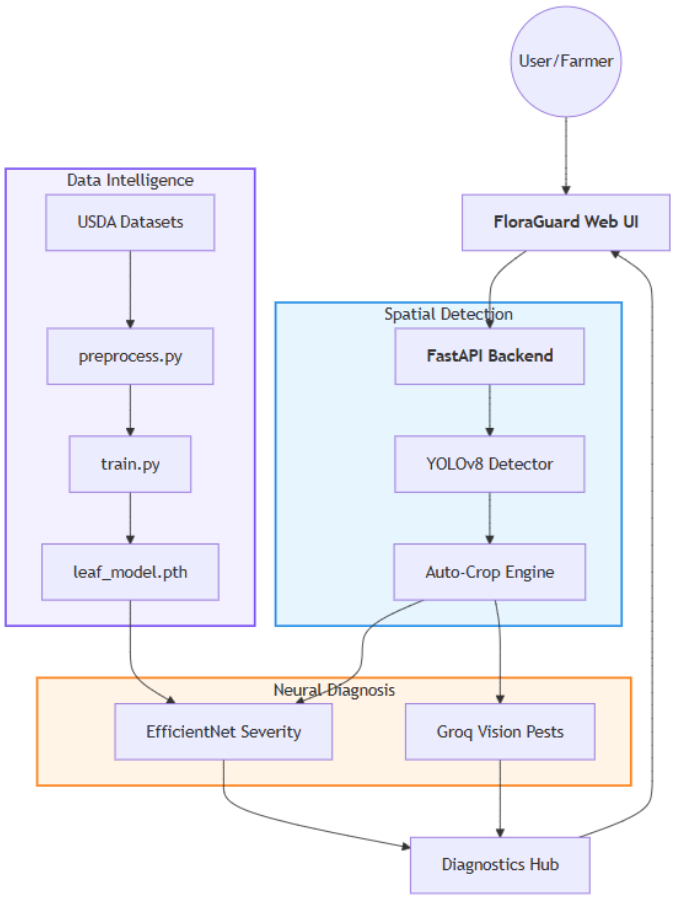



Fig. 1. Overall System Architecture

## V. Methodology

Methodology consists of a structured multi-step pipeline for plant leaf disease detection and assessment. It consists of data preparation, spatial detection, region-oriented processing, and classification. The proposed system is created to maintain robustness across many different input datasets. This is done by using a standardised set of preprocessing techniques in conjunction with advanced deep learning models. The methodology focuses on refining the input data sequentially to enhance feature quality and predictive accuracy.

### A. Data Collection

To ensure diversity and generalization when training models, a range of publically available agriculture datasets is used by the system. The principal data sets used are the PlantVillage data set, which contains labeled photos of diseased and healthy plant leaves that were taken under controlled environments; and the PlantWild data set that has real world photos of plants taken under different natural environments including different types of lighting, occlusion and diverse backgrounds. The ability of the model to learn both patterns in structure and patterns in randomness will give it the robustness necessary for practical use. The importance of using diverse datasets to prevent overfitting and enhance model generalization has been well documented in recent literature. [1], [5].

### B. Data Preprocessing

Having quality input data helps increase model performance through model improvement processes. Collecting quality data is important in that it helps create datasets of uniform images and eliminates unnecessary data. Before using images for training, the images must be preprocessed to increase probability of being compared with other images when measuring outcome. Next, the images must be cleaned so the training does not fail due to low-quality images. Model performance will be improved if the number of images available for training increase than the total number of images available for use in training.

Normalization is used to adjust pixel values so they are all on an equal range and help to provide consistent convergence while training a model. In deep learning this is especially beneficial because it helps alleviate the effects of different lighting conditions on the variations in pixel values. Also, formatting procedures are performed to properly organize the data base into the training and validation before the model can be run for the purposes of evaluation.

Rotation, flipping, and scaling, some augmentation techniques that simulate the real world to help apply more variance to the model and help generalize better on data that was never presented to the model in its training set, further improve the robustness of the model.

### C. Model Design

The multi-stage design model's architecture consists of detection and classification components that are separated into multiple tiers. The first tier is made up of the object-detection module using the YOLO architecture. This module finds and locates disease-affected areas inside an input image, and YOLO was chosen because of its high accuracy rate and ability to provide real-time results. [7], [8].

**Multi-Task Learning Loss**

$$L_{total} = L_{det} + \lambda L_{cls} \tag{2}$$

where

- $L_{det}$ = YOLO detection loss
- $L_{cls}$ = EfficientNet classification loss
- $\lambda$ = balancing parameter

In the second phase of the extraction process, a region extraction system extracts all of the isolated regions so there is only an area of interest within the image to perform any further analysis. This greatly improves the quality of any features because it eliminates any background noise.

The last module is a classification module that uses EfficientNet architecture to predict disease type and severity. EfficientNet is the chosen model for its optimized scaling strategy, and EfficientNet has superior performance levels compared to all other choices in image classification tasks

[3], [9]. The ability to integrate both detection and classification models allows the system to capture both spatial and semantic information thus increasing the overall accuracy of the diagnosis.

The outputs of each stage are aggregated using a diagnostic aggregation procedure to produce a single output that contains both localization and classification information.

### *D. Feature Extraction*

Deep learning models provide implicit feature extraction, as they automatically generate hierarchical representations of input data. During the detection phase, convolutional layers extract spatial features (e.g., boundaries of regions, object localization) used to determine the location of any infected areas on an image of a leaf.

**Convolution Feature Extraction**

$$F_k(x,y) = \sum_{i=1}^{m}\sum_{j=1}^{n} K_k(i,j)\cdot I(x+i, y+j) + b_k \quad (3)$$

where

- $F_k$ = $k^{th}$ feature map
- $K_k$ = convolution kernel
- $b_k$ = bias

The Classification stage involves the extraction of characteristics of diseases from those high-order features such as changes in colour, texture and distortion of structures that allow for differentiation between different disease classes and degrees of severity.

Utilizing a combination of spatial and semantic extraction of features provides greater accuracy and interpretability by utilizing a combination of multitasked level feature learning through modern deep learning methods for plant disease detection using images. [4], [13].

TABLE III
EXTRACTED FEATURES

| **Feature** | **Description** |
|---|---|
| Color Features | Pixel intensity variations indicating discoloration |
| Texture Features | Local patterns such as spots, lesions, and surface roughness |
| Edge Features | Boundaries of infected regions and irregular structural patterns |
| Shape Features | Structural deformation and morphological changes in leaf geometry |
| Spatial Features | Location and distribution of infected regions across the leaf |
| Area Features | Proportion of affected region relative to total leaf area |
| Severity Indicators | Degree of infection based on spread, density, and intensity |

## VI. IMPLEMENTATION DETAILS

The proposal involves a modular real-time inference system that combines a deep learning algorithm with an API for efficient plant disease diagnosis. Low latency processing, scalability, and smooth interactions between components are the key design considerations for this system. Each step in the process (feeder operation, pre-processing, detection, classification and final result generation) will be developed as individual components using a service-oriented architecture.

A lightweight RESTful API Framework Develops the Backend to Allow Core Communication between the User Interface and the Model Pipeline (i.e., the processing unit) via API Endpoints That Transmit Input Images from the User Interface to the API endpoints for Processing & Validation and Preprocessing prior to Inference Modules Receiving the Images. This Design Provides Both Flexibility and Compatibility Across Multiple Deployment Environments such as Web-Based Systems and Edge Systems.

Infected Areas are located through a process called Detection. To do this, a Real Time infection detection method uses a YOLO-based Detection Method for the localization of the location of Infected Areas. The YOLO Architecture has been established as a leading objective detection method [7], [8], and new architectures like YOLOv8 also provide higher accuracy for agricultural detection applications [2]. The Detection Method outputs bounding boxes around infected areas with a probability score which is later filtered for low confidence probabilities using a filter based on predetermined thresholds.

Once the detected areas of interest have been identified, a mechanism will allow for the extraction of regions associated with each of those areas, so that only these regions—i.e., those that relate to diseases—are used in future processing. This step removes any background interference and enhances the quality of features in comparison to traditional classification models that operate over the entire image space. [11], [13].

The classification phase uses a convolution neural network based on EfficientNet which is known for its scaled efficiency and for being a high performing model for image classification applications [3], [9] . Using transfer learning, the EfficientNet Model is trained with pre-trained features and adjusted appropriately for agricultural data sets in order to generate class probabilities to assist with confidence decision making [6], [17].

The second (or intermediate) evaluation layer receives output from each of the previous detection and classification layers. Confirmation of prediction reliability takes place through the review of confidence levels and consistency checks; therefore, predicted severity of disease is representative of the size of the areas detected as infected, i.e., logical consistency of final output. Compared to the approach that uses only one model, this process provides greater assurance of prediction reliability using multiple stages of validation. [1], [5].

In addition to a reasoning module that allows the interpretation of model outputs and generation of structured diagnostic insights, there is an added benefit of mapping the predicted output of the model to domain-related knowns to improve the interpretability of the model and support real world decision making for agricultural applications.

The system has been designed to load models and utilize them effectively. The system loads models once and then repeatedly uses them to save time as well. A computer can utilize a specific component known as a graphics processing unit (GPU) to work at a faster rate. If your computer is weak or not very powerful, it can run on CPU as well. The system has also been designed using flexibility so future ideas or models will be able to run; examples include joint use of convolutional and transformer model types. [4].

TABLE IV
TECHNOLOGY STACK

| Layer | Component | Implementation |
|---|---|---|
| Application Layer | Backend Interface | FastAPI-based REST API for request handling and pipeline orchestration |
| Data Processing Layer | Image Preprocessing | OpenCV and NumPy for resizing, normalization, and transformation |
| Detection Layer | Object Detection Model | YOLOv8 for real-time localization of infected regions |
| Classification Layer | Disease Classification Model | EfficientNet architecture for feature learning and severity prediction |
| Learning Framework | Model Development | PyTorch for training, optimization, and inference |
| Training Strategy | Model Optimization | Transfer learning with fine-tuned pre-trained weights |
| Data Layer | Dataset Sources | PlantVillage (controlled) and PlantWild (real-world) datasets |
| Inference Layer | Execution Engine | PyTorch runtime for model inference |
| Deployment Layer | System Deployment | Configurable deployment across local, server, and edge environments |

### A. System Workflow

The system workflow consists of an ordered container for processing raw image inputs and transforming them into trustable diagnosis outputs via a series of processing phases (sequential processes). Each phase has an intended purpose of improving the input data through repeated processing so that the resulting features are of high quality, making accurate predictions possible and resulting in efficient inference.

**Input Acquisition:** The process of getting an image of a leaf starts with uploading it via a user interface. After being uploaded, the image is sent to the backend server via an API call, where subsequent checks will be made on the base file type, resolution and integrity of the file.

**Preprocessing:** The acquired image undergoes preprocessing operations, including resizing to a fixed resolution, normalization of pixel values, and conversion into tensor format. These steps standardize the input data and reduce variability caused by environmental factors such as illumination changes and background noise.

**Detection:** We send the prepared image through the part of the system that will search for diseases. The tool used by this part of the system to identify and indicate where the infection is found on the image is called YOLO. When YOLO determines an infection's location, it draws a box around the area on the image, gives a probability score of the likelihood of infection at that area, and identifies which areas do not have a probability score (indicating we can trust YOLO's determination of that area as an infection site). YOLO is excellent at identifying infection locations within the image.

**Region Extraction:** The system uses the coordinates of the bounding boxes to select the areas on the images that are related to infection. This means that we can remove the background, where there are no disease-related areas of the image, and focus on those regions that have clarity of features, with less interference from any background that is not related to the disease itself.

**Classification:** An EfficientNet-based classification algorithm will process the regions extracted from the images and will provide predictions on the disease classification as well as determines how severe it may be. This classification data is further utilized to create probability distributions for each of their respective disease classes and will allow for clinical decision-making through utilizing probability distributions and therefore improving diagnostic accuracy.

**Evaluation and Aggregation:** At this point, detection and classification output are consolidated for a final output prediction that is consistent and reliable. Confidence thresholds can be used, while spatial characteristics (in this case infected area size) can be correlated with severity predictions to establish logical consistency among predictions. Validation in this manner contributes to increase robustness and the elimination of false positives.

**Output Generation:** The results from this stage are packaged into a structured output that includes all the calculated metrics and summarizes the total number of areas identified for investigation, a corresponding predicted diagnosis with associated severity level and associated level of certainty. The results will be sent to the user interface as well as elaborate on the nature of the findings in order to give clear and actionable information related to the diagnosis.

## VII. EXPERIMENTAL SETUP

By utilizing both an experimental and real-world environment to evaluate the performance of the newly developed multi-phase diagnostic for detecting plant diseases, the centre looks at detection accuracy, classification performance and overall reliability of prediction. Both the testing experiment will use various agricultural datasets from standard databases and use the commonly used measures in computer vision and deep learning methods.

### A. Dataset Description

The evaluation of the proposed system is conducted using two complementary datasets to ensure both robustness and generalization capability.

**PlantVillage Dataset:** The PlantVillage dataset is utilized as a key benchmark dataset. It contains a large volume of accurate and labelled pictures of plants taken in a consistent environment and have a uniform background (i.e., they are all

in the same lighting conditions, and are photographed at the same angle). Having this kind of dataset means the model can learn clear and well-defined disease patterns. The dataset is therefore appropriate for initial training and validation. [11], [12].

**PlantWild Dataset:** The PlantWild dataset has been used for evaluation purposes in order to test real-world application performance of the computer vision system. The dataset consists of images taken in an uncontrolled natural environment. This includes variability in lighting, background complexity, occlusion, and leaf orientation. The addition of this variability produces real-world challenges that allow the computer vision system to be evaluated for generalization and robustness in practical use. [1], [5].

**Data Splitting and Augmentation:** There are three subparts to the dataset; The Training set is for learning, and then validating by adjusting the settings in the validation set. Finally we use the Test set to determine the actual performance of the model. Moreover, a technique to make the data exciting; is to rotate, flip and change the size of it either bigger or smaller which will help the model learn more about it's job from the available data.

### B. Evaluation Metrics

TThe proposed model is checked to see how well it works. This is done by using methods to test its ability to classify and detect things. To check how well the model classifies we look at how accurate it's how precise it is how well it recalls things and its F1-score. These things help us understand if the model can make predictions and if it has a good balance, between false positives and false negatives. The proposed model is also checked to see how well it detects things and this is done by looking at the IoU and the mAP of the proposed model.

**Severity Formula:**

$$Severity(\%) = \frac{Area_{infected}}{Area_{leaf}} \times 100 \tag{4}$$

TABLE V
EVALUATION METRICS

| Metric | Formula | Purpose |
|---|---|---|
| Accuracy | $(TP+TN)/(TP+TN+FP+FN)$ | Overall correctness of predictions |
| Precision | $TP/(TP+FP)$ | Measures proportion of false positives |
| Recall | $TP/(TP+FN)$ | Measures detection completeness |
| F1-Score | $2PR/(P+R)$ | Balances precision and recall |
| IoU | Overlap / Union | Measures localization accuracy |
| mAP | Mean(AP) | Evaluates overall detection performance |

## VIII. RESULTS AND ANALYSIS

To determine if the proposed approach performs getthedisease diagnosed and detected at the regional level and produce better prediction reliability by concentrating on the area of interest than just an image of the area of interest. The results produced were obtained from a series of experiments with various patients using the proposed method and indicate that through the use of a system with multiple stages, improved performance is achieved as compared to traditional systems that were used as single-stage systems.

### A. Performance Results

To evaluate the classification, four popular metrics: accuracy, precision, recall, and F1-score, are used. In comparison to baseline CNN models, EfficientNet based models exhibit superior classification performance. This is primarily attributed to improved features being represented and a more efficient way of scaling. Further improvement can be attributed to the use of a region focused processing pipeline where classifications are made at the level of extracted regions instead of entire images thus reducing the ability to confuse the classifications due to the presence of backgrounds.

TABLE VI
MODEL PERFORMANCE

| Model | Accuracy | Precision | Recall | F1-Score |
|---|---|---|---|---|
| Baseline CNN | 0.88 | 0.86 | 0.85 | 0.85 |
| Transfer Learning Model | 0.91 | 0.90 | 0.89 | 0.89 |
| Proposed (EfficientNet + Detection) | 0.94 | 0.93 | 0.92 | 0.92 |

### B. Comparative Analysis

The comparative assessment performed compares the various methods available for identifying diseases affecting plants. Image processing systems based on traditional methods show poor identification performance due to their inability of being able to account for changes within the surrounding environment. Methods that are based on Convolutional Neural Networks provide greater accuracy than traditional methods; however, provide little spatial awareness. Detection-based systems allow for some form of localization, but are not able to fully capture the features associated with different plant diseases when utilized on their own.

Combining detection with classification creates a single system; both spatial analysis and semantics can be performed using this system. Because of this combination, the prediction results will now be better than before, especially under conditions of complexity, such as varying amounts of background noise and variability that are present in real-world scenarios.

TABLE VII
COMPARISON WITH EXISTING SYSTEMS

| Approach | Methodology | Accuracy | Limitation |
|---|---|---|---|
| Traditional Methods | Feature-based analysis | 0.72–0.80 | Sensitive to noise and illumination |
| CNN-based Models | Image-level classification | 0.85–0.90 | No localization capability |
| Detection-based Models | Object detection (YOLO) | 0.88–0.91 | Limited classification depth |
| Proposed System | Detection + Classification | 0.94 | – |

### C. Observations

Detection of Regions Ahead of Classifying Helps Model Performance by Improving Feature Relevance to Model Output; In Addition, The Training of this Model on Real World Data Allows it to be More Robust to Environmental Variances in Terms of Region Where Detection Occurred. Also, The Similarity Between Detection Output and Classifier Output Issues Reduces the Number of False Classifications, Demonstrating the Benefit of A Multi-Stage Model Design. Therefore, The System As A Whole Provides An Equal Improvement In The Accuracy And Reliability Of The Models, And Is Ideal For Use In An Agricultural Environment.

## IX. Discussion

Experimental findings indicate that the proposed multi-stage framework enhances diagnostic performance through a combination of spatial detection and classification. By first localizing the area of the affected area before performing a classification, it will enable feature extraction to focus on the relevant sections of the image, thus avoiding any background noise and therefore improving the prediction results. The region-aware (localised) approach provides a direct resolution to one of the key limitations that are experienced with standard (classical) CNN models, which work with whole images without accounting for spatial relationships [6], [18]. The overall increase in accuracy provides strong evidence of the effectiveness of integrating detection with classification in a single pipeline.

Detection and classification blocks are complementary to one another. The detection block accurately locates infected areas, while the classification block extracts high-level semantic features to improve the accuracy of disease diagnosis [7], [8]. These two blocks work together to create both spatial and semantic awareness of plant diseases [3], [9]. The third evaluation block also works to enhance the reliability of predictions by filtering out low reliability predictions and aligning predicted severity of infection with the actual infected area; this follows recent trends in deep learning integration [1], [5].

## X. Future Scope

Future studies might focus on implementing advanced deep learning structures (transformer-style visual models or hybrid architectures made up of convolutional neural networks and transformers) to the system. Models with these types of architectures seem to perform better than existing architectures in establishing complex feature relationships [4]. Future research will also incorporate multi-modal inputs, such as environmental conditions (e.g., temp, humidity, and soil), which will help improve diagnostic accuracy by providing additional disease/pest contextual information, apart from visual characteristics of disease/pest features.

The system may be adapted to allow for large-scale agricultural use by expanding its cloud infrastructure and distributed processing capabilities from a deployment standpoint. By using optimized models in an edge deployment, it is now possible to offer real-time diagnostic capabilities in resource-constrained settings, consistent with recent trends toward using edge AI for agricultural systems [2]. In addition, developing a dataset to include broader ranges of crop types and disease categories will enhance the generalization of models, and using explainable AI methods will improve users' understanding of and confidence in the predictions made.

## XI. Conclusion

Using a unified pipeline to combine detection of object to classification with in one pipeline. The proposed approach has resolved some of the key limitations found in the current methods. By using regional analysis, the proposed methodology allows the system to concentrate on regions with diseased tissue rather than having to process the entire image. The use of both YOLO for localisation of an object and EfficientNet for accurate classification, provides improved overall diagnosis performance base on both location and types of disease.

Results from the experiment demonstrated that the proposed system meets all criteria for accuracy and reliability per all measures of evaluation used to assess it. In combination with detection and classification, they contribute to the improvement of prediction stability by diminishing the impact of noisy background on predictions and enhancing the significance of features. Additionally, our results provide evidence of the generalization properties of the system, since it was able to accurately analyze both artificial and natural samples. These factors indicate that the system is a viable tool for agricultural use in a practical setting.

Farmers can identify agricultural disease outbreaks prior to them causing significant agricultural damage by using our agricultural monitoring systems. You'll gain quicker, dependable, easy-to-use forecasts to produce improved annual crop production and develop better cropping practices using the information provided by these systems.

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